POLITEKNIK MALAYSIA
TUN SYED NASIR



CHAPTER 4: WASTEWATER TREATMENT | Ts. MUHAMMAD AZRI IZANI BIN MOHAMAD HALIM

TREATMENT METHOD

4 main types of wastewater treatment used to remove pollutants from wastewater:

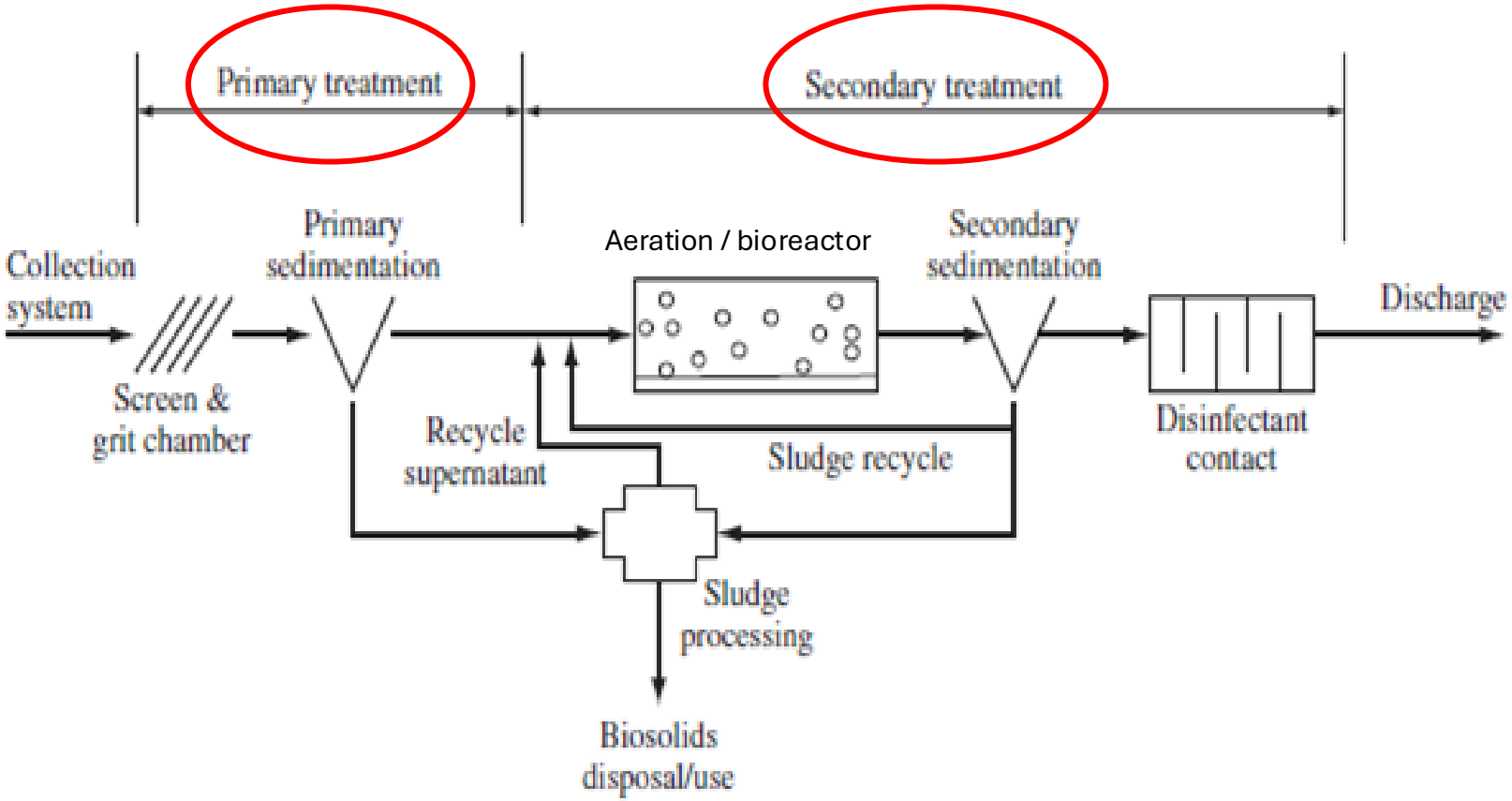
1. Physical treatment
2. Chemical treatment.
3. Biological treatment.
4. Hybrid processes.

TREATMENT METHOD

Treatment Category	Process	Function
Physical Treatment	1. Membrane filtration 2. Adsorption, Dissolved Air Flotation (DAF) 3. Evaporation	Removes suspended solids, oil, grease, and some dissolved pollutants using physical separation.
Chemical Treatment	1. Coagulation 2. Flocculation 3. Photocatalysis	Uses chemicals or chemical reactions to destabilize and remove contaminants.
Biological Treatment	1. Anaerobic digestion 2. Aerobic digestion	Uses microorganisms to degrade organic matter and reduce BOD/COD.
Hybrid Processes	1. Membrane bioreactor (MBR) 2. Microbubble flotation + coagulation 3. Photofermentation	Combines two or more treatment methods to improve efficiency and achieve higher-quality effluent.

WASTEWATER TREATMENT PROCESS

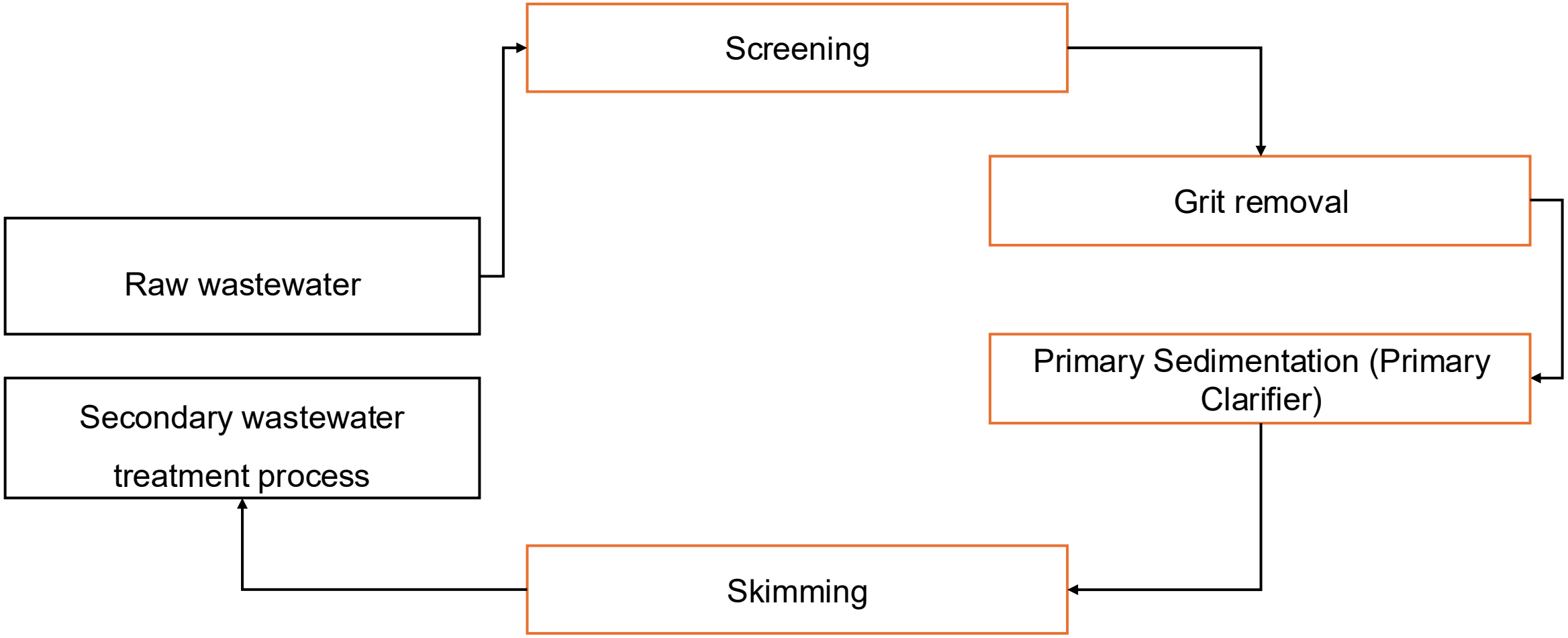
- This is the typical wastewater treatment process providing primary and secondary treatment.



PRIMARY WASTEWATER TREATMENT

- Primary wastewater treatment uses physical processes such as screening, grit removal, sedimentation, and skimming to remove settleable and floatable solids.
- It typically removes 50% to 70% of TSS and 25% to 40% of BOD before wastewater enters secondary treatment.
- The main objective is to eliminate large debris, settleable solids, floatable materials (e.g., oil, grease, and scum), and other pollutants that cannot pass through screening devices, thereby reducing the pollutant load before subsequent biological treatment processes.

PRIMARY WASTEWATER TREATMENT



EXAMPLE 1

1. State 4 types of wastewater treatment.

2. State 4 process involves in the primary wastewater treatment.

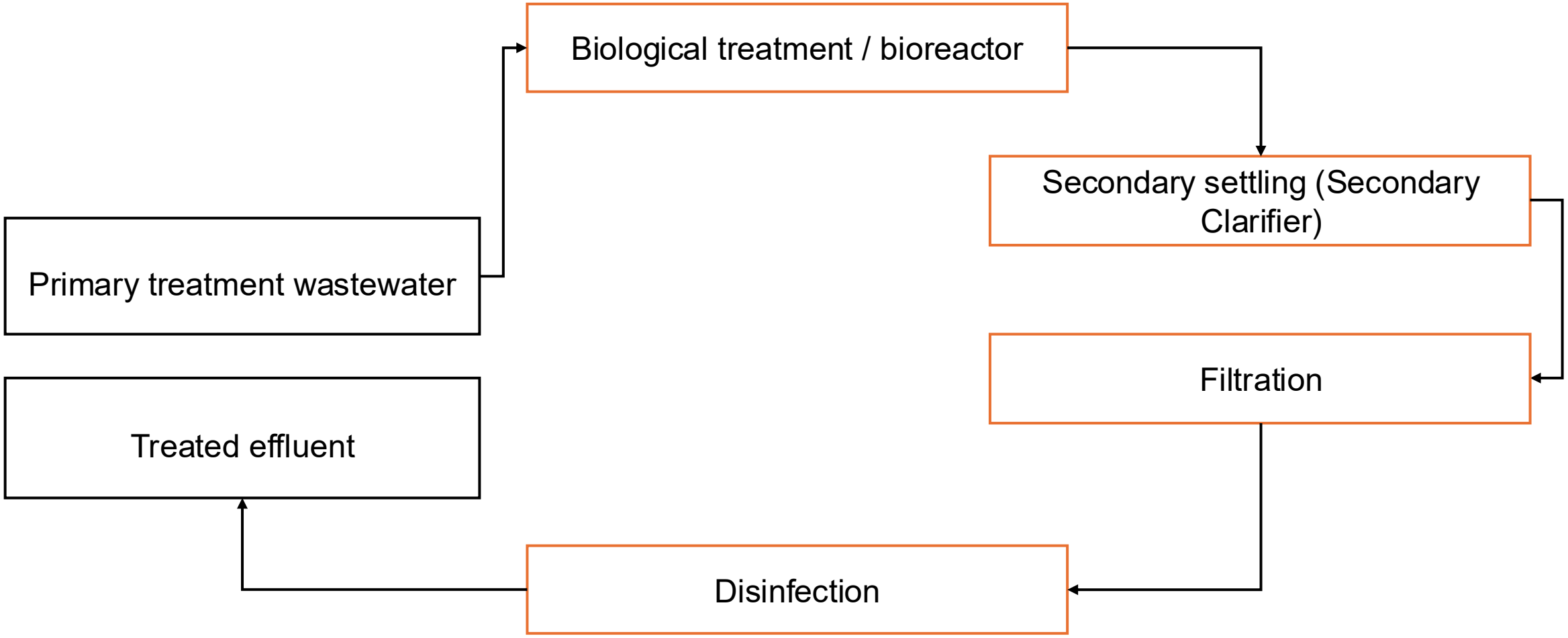
PRIMARY WASTEWATER TREATMENT

Process	Function	Materials Removed
Screening	Removes large debris using bar screens	Plastics, rags, wood, leaves
Grit Removal	Slows flow to allow heavy inorganic particles to settle	Sand, gravel, grit, eggshells
Primary Sedimentation (Primary Clarifier)	Allows suspended solids to settle by gravity	Organic solids, sludge
Skimming	Removes floating materials from the surface	Oil, grease, fats, scum

SECONDARY WASTEWATER TREATMENT

- Secondary wastewater treatment is designed to achieve further removal of biochemical oxygen demand (BOD) and suspended solids beyond the levels attainable through primary treatment and sedimentation.
- A properly designed and operated secondary treatment system can typically remove up to 90% of BOD and 90% of suspended solids, significantly improving effluent quality.
- The process primarily relies on biological treatment, where microorganisms metabolize and convert organic pollutants present in wastewater into stable, low-energy compounds such as carbon dioxide, water, and biomass.

SECONDARY WASTEWATER TREATMENT



SECONDARY WASTEWATER TREATMENT

Process	Function
Bioreactor	Biological degradation of organic matter by microorganisms
Secondary Clarifier	Settling and removal of biological flocs
Filtration	Removal of remaining fine suspended particles
Disinfection	Inactivation of pathogenic microorganisms

EXAMPLE 2

- State the function of process involving bioreactor, secondary clarifier, filtration and disinfection.

WASTEWATER RELATED TO O&F INDUSTRY (POME)

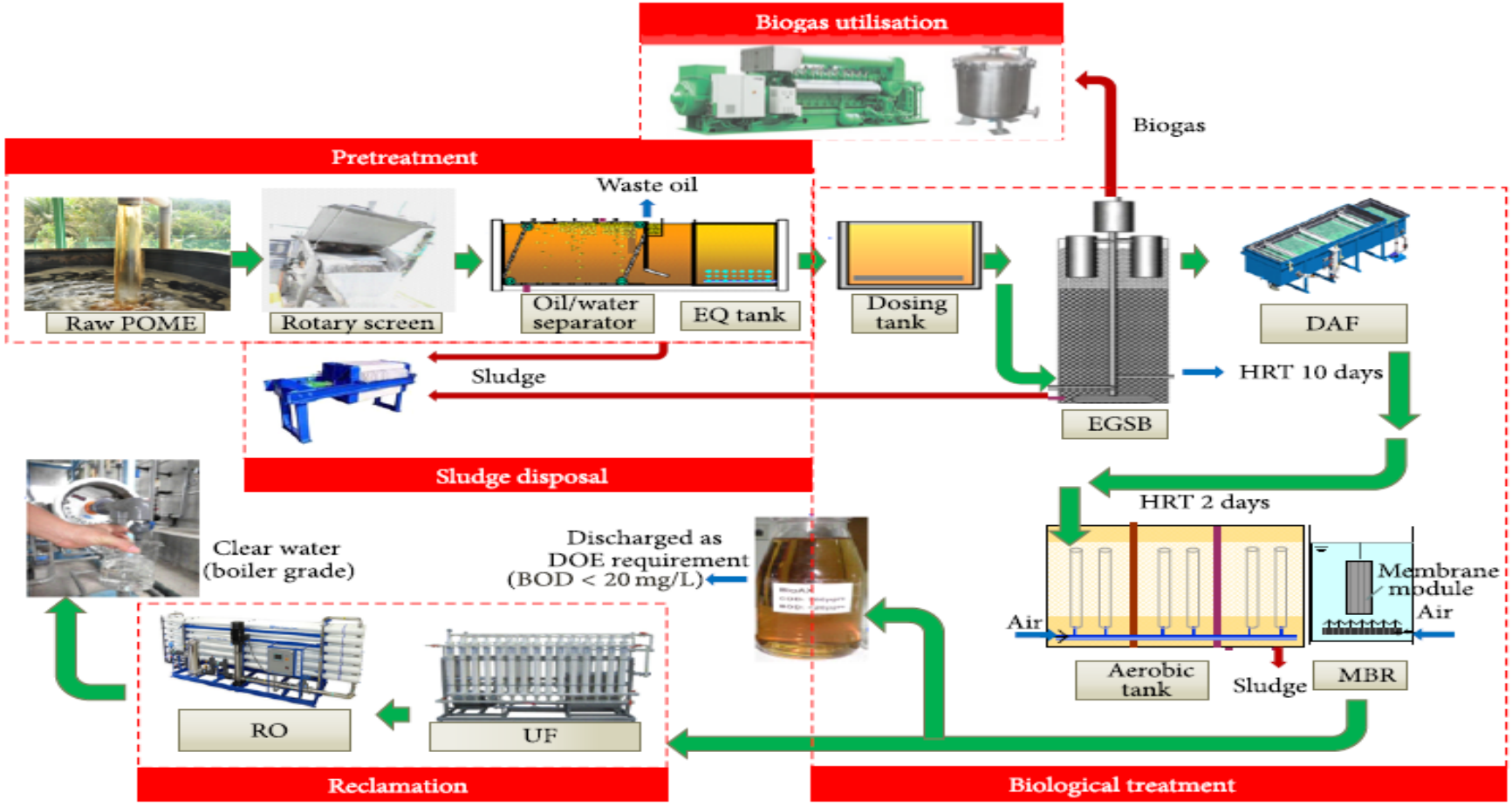
Palm Oil Mill Effluent (POME) is the liquid waste generated during the processing of fresh fruit bunches (FFB) in the palm oil industry.

It is one of the major industrial wastewater streams in Malaysia and other palm oil-producing countries.

POME is produced from several milling operations:

1. Sterilization of fresh fruit bunches (steam condensate + wastewater)
2. Clarification of crude palm oil
3. Hydrocyclone and separator discharge
4. Washing and cleaning processes in the mill

WASTEWATER RELATED TO O&F INDUSTRY (POME)



WASTEWATER TREATMENT TECHNIQUE

- There are several techniques that suitable to treat wastewater according to their characteristics and pollution levels:
    1. Physico-chemical treatment methods.
    2. Trickling Filter (Biological Filtration).
    3. Nutrient Removal (Nitrogen & Phosphorus Removal).
    4. Activated Sludge Process (ASP).
    5. Oxidation Pond (Waste Stabilization Pond).

PHYSICO-CHEMICAL TREATMENT*

- There are 10 processes involve in physico-chemical wastewater treatment:

1. Screening: Removal of large debris (rags, plastics, sticks)
2. Grit removal (grit chamber): Separation of sand, gravel, and heavy inorganic particles
3. Primary sedimentation (primary settling tank): Settling of suspended solids by gravity
4. Oil-water separation (gravity/API separator): Removal of free oil and grease
5. Coagulation: Chemical destabilization of fine particles and emulsions

PHYSICO-CHEMICAL TREATMENT

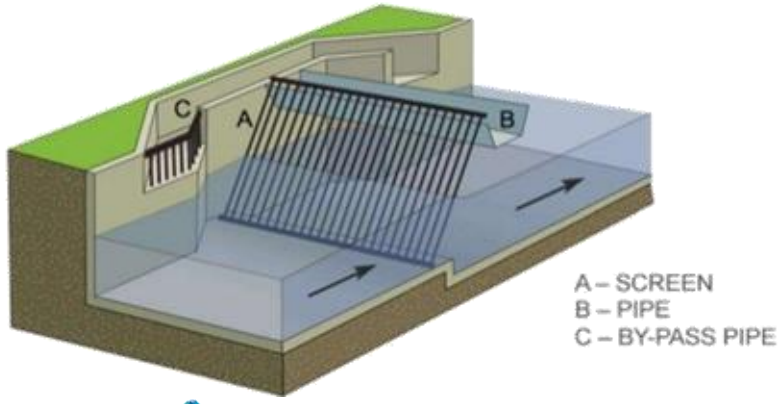
- There are 10 processes involve in physico-chemical wastewater treatment (CONTINUE):

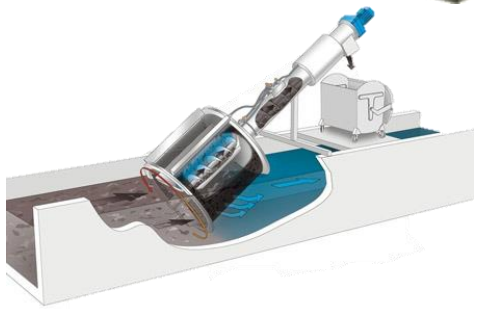
6. Flocculation: Destabilized particles into larger flocs
7. Chemical precipitation: Removal of dissolved contaminants (e.g., metals, phosphorus)
8. pH adjustment (neutralization): Control of acidity/alkalinity using acids or alkalis
9. Equalization: Balancing flow rate and pollutant concentration (BOD, TSS, pH)
10. Dissolved Air Flotation (DAF): Removal of oil, grease, and fine suspended solids using air bubbles

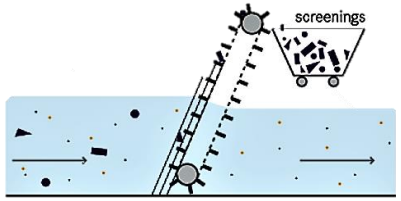
PHYSICO-CHEMICAL (SCREENING)

- Screening is the first step in wastewater treatment. It removes large floating materials that can damage equipment or block pipes.
- Screening process working principles:

1. Wastewater flows through the screen bars.
2. Large solids are trapped on the screen surface.
3. Collected materials are removed periodically.
4. In large plants, at least two parallel screens are used.
5. If only one screen is installed, a bypass channel is required.



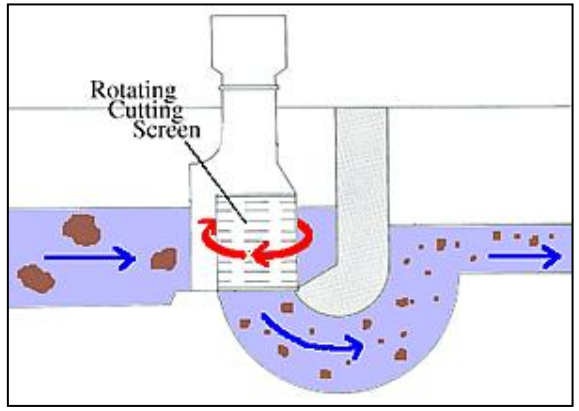






PHYSICO-CHEMICAL (COMMINUTOR)

- A comminutor is a device used to grind or shred large solid materials in wastewater into smaller particles so they can pass through the system without clogging pipes or damaging equipment.



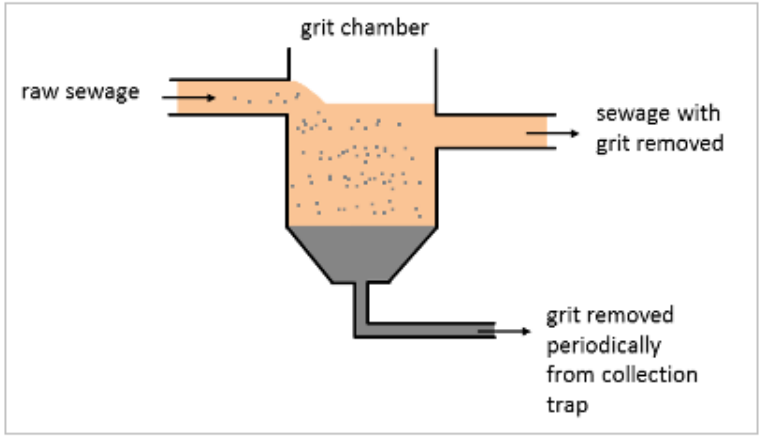
- Comminutor working principles:

1. Solids are shredded while water passes through
2. Reduced particles move to downstream units (e.g., sedimentation tank)
3. These grinders consist of fixed, rotating or oscillating teeth or blades, acting together to reduce the solids to a size which it can pass through an openings of about 6 mm.



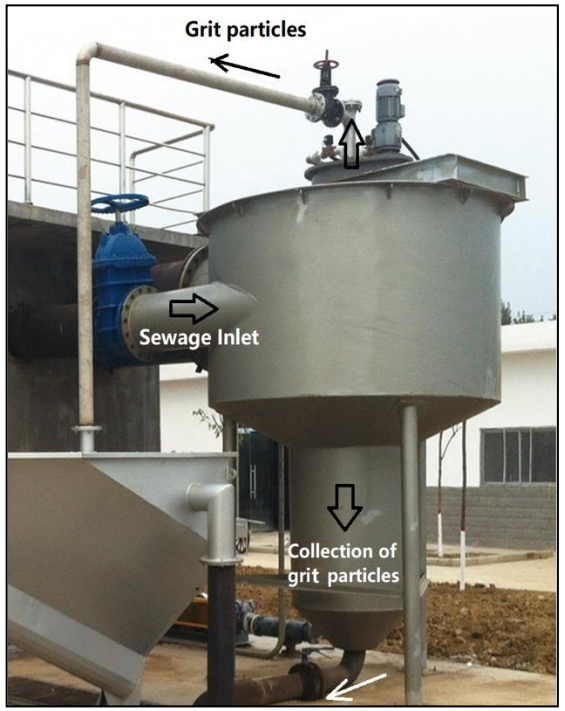
PHYSICO-CHEMICAL (GRIT CHAMBER)*

- A Grit chambers are designed to separate heavy inorganic particles from organic matter based on differences in settling velocity.



- Grit chambers working principles:

1. Wastewater enters the grit chamber.
2. Flow velocity is reduced and controlled.
3. Grit (sand, gravel) settles to the bottom.
4. Organic matter continues to flow forward.
5. Settled grit is collected and removed periodically.



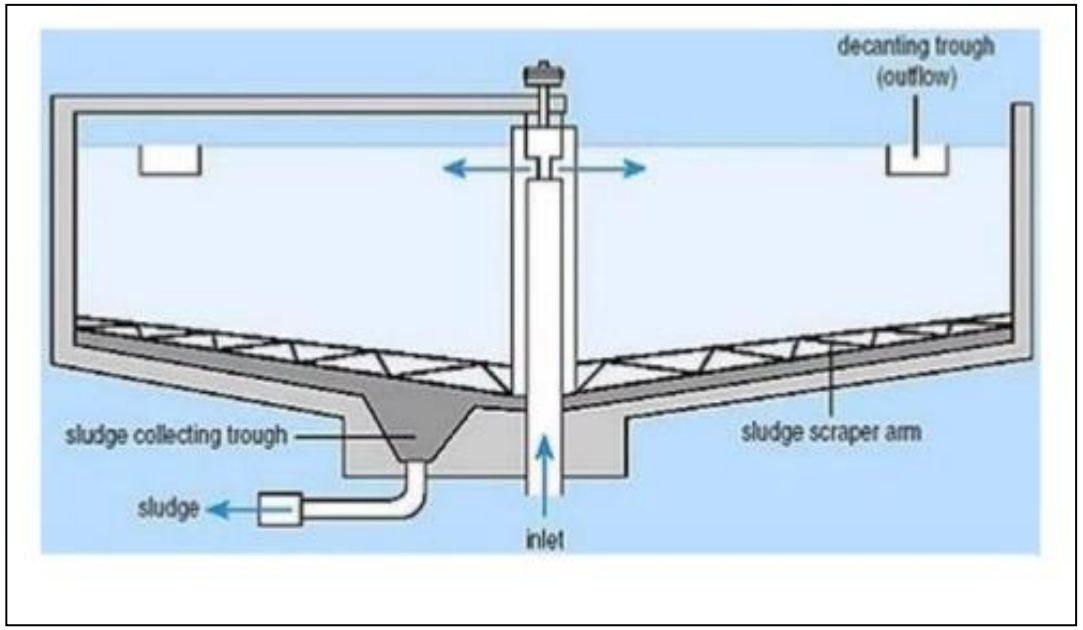
EXAMPLE 3

- State 4 other wastewater techniques other than physico-chemical treatment method.
- Explain grit chamber working principles.

PHYSICO-CHEMICAL (PRIMARY SETTLING TANK)

- A primary settling tank, also known as a primary clarifier or sedimentation basin, is a key unit in wastewater treatment that follows preliminary processes such as screening and grit removal.
- It is designed to reduce the velocity of wastewater flow so that suspended solids can settle out by gravity under controlled conditions.





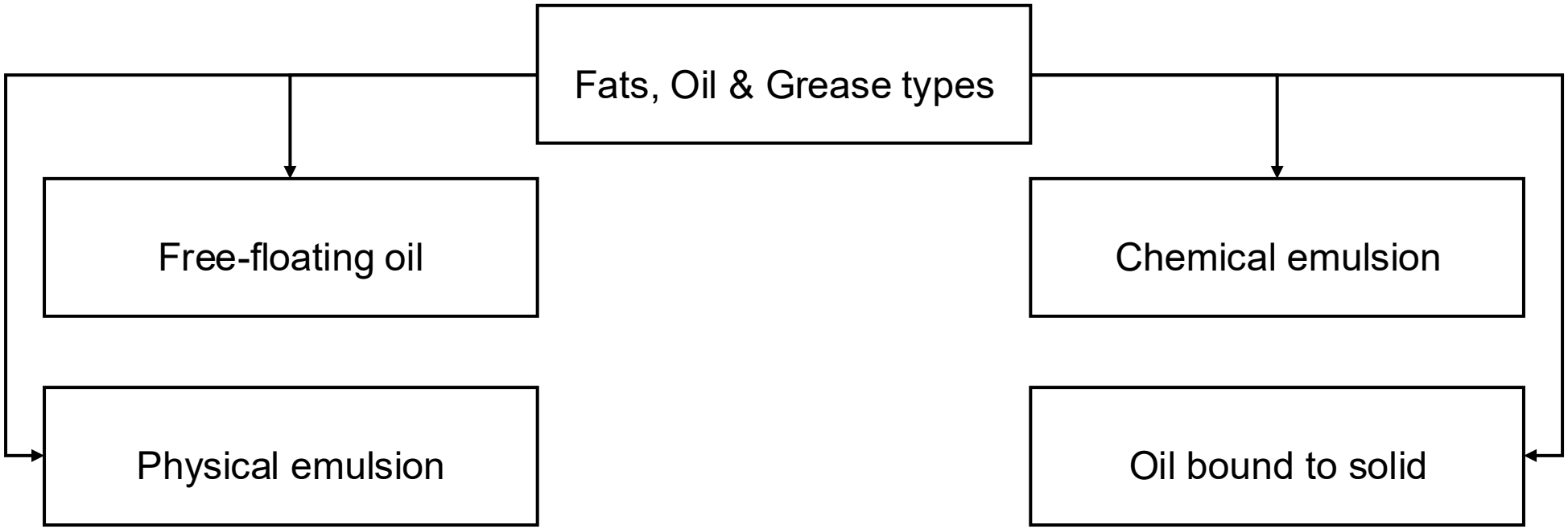
PHYSICO-CHEMICAL (PRIMARY SETTLING TANK)

- Working principles of primary settling tank:

1. Wastewater from preliminary treatment enters the primary settling tank (also called sedimentation basin or primary clarifier)
2. Flow velocity is reduced to create calm conditions that allow gravity settling
3. Heavier suspended solids settle to the bottom forming **primary sludge (raw sludge)**
4. Lighter materials such as oil, grease, and scum float to the surface and are removed
5. Clarified water flows out from the tank for further treatment or disinfection

PHYSICO-CHEMICAL (OIL-WATER SEPARATION)*

- Fats, oil, and grease (FOG) are common pollutants in wastewater and may exist in different physical forms such as free oil, emulsified oil, or oil bound to solids.



PHYSICO-CHEMICAL (OIL-WATER SEPARATION)

- Fats, oil, and grease (FOG) separation principles:

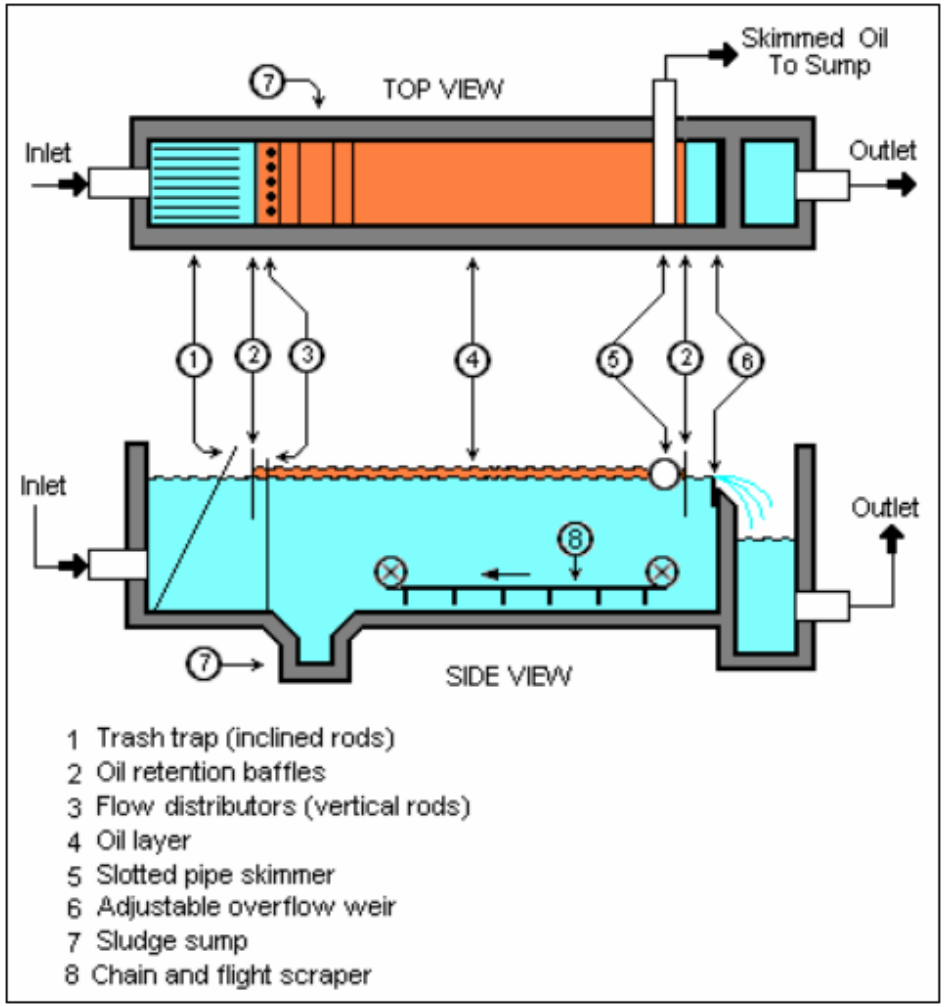
Type of Fats Oil & Grease	Separation Principle	Mechanism
Free-floating oil	Gravity separation	Oil has lower density than water, so it rises to the surface
Physical emulsion	Thermal + chemical destabilization	• Heat reduces viscosity • coagulants break weakly bound oil-water mixture
Chemical emulsion	Chemical breaking of stabilizing agents	Emulsifying agents are neutralized or destroyed to release oil droplets
Oil bound to solids	Solid-liquid separation + flotation/sedimentation	Oil attached to particles is removed together with suspended solids

API SEPARATOR

- An API (American Petroleum Institute) Separator is a gravity-based oil-water separation unit used to remove free oil, grease, and suspended solids from industrial wastewater.
- Working Principle of API separator:
    1. Wastewater enters the API separator.
    2. Flow velocity is reduced to create calm (quiescent) conditions.
    3. Heavy suspended solids settle to the bottom.
    4. Free oil droplets rise to the surface.
    5. Water remains in the middle layer.
    6. Surface oil is skimmed off.
    7. Settled sludge with higher density that settle downward is scraped and removed.
    8. Clarified wastewater is sent for further treatment (e.g., DAF or biological treatment).

API SEPARATOR

- An API (American Petroleum Institute) schematic diagram:



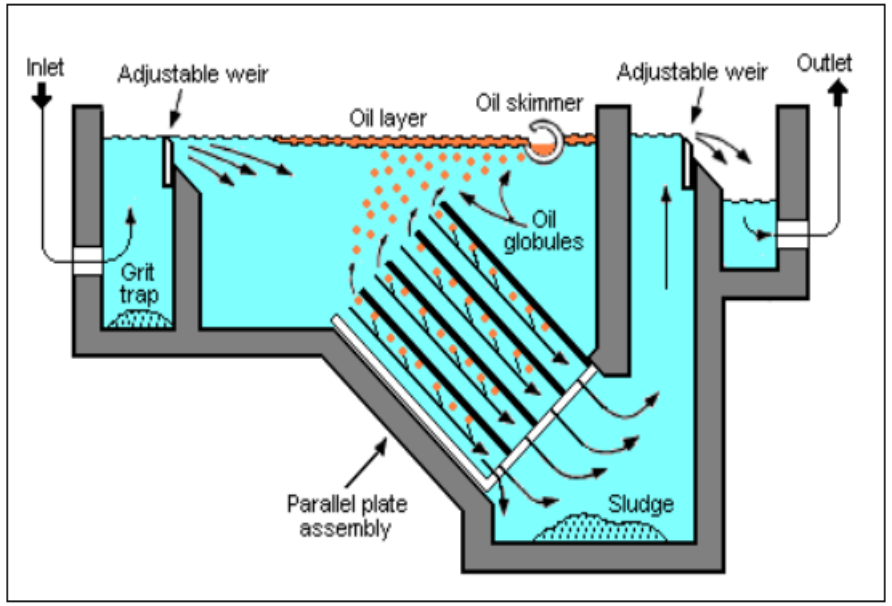
EXAMPLE 4

1. State 3 types of fats, oil and grease form including their separation principles.

2. What is API separator?

COALESCER

- A coalescer is an improved version of an API separator that uses inclined (tilted) parallel plates to enhance oil-water separation.
- Like an API separator, it relies on the difference in density (specific gravity) between oil and water. However, the parallel plates provide additional surfaces where small oil droplets can combine (coalesce) into larger droplets, making them easier to separate by gravity.



Feature	API Separator	Coalescer
Separation mechanism	Gravity only	Gravity + coalescence
Plate packs	No	Yes
Space requirement	Large	Small
Oil removal efficiency	Moderate	Higher
Small droplet removal	Limited	Better
Capital cost	Lower	Higher

COALESCER

- Working Principle of API separator:
    1. Oily wastewater enters the coalescer.
    2. Wastewater flows through a series of tilted parallel plates.
    3. Small oil droplets contact the underside of the plates.
    4. Oil droplets merge (coalesce) into larger oil globules.
    5. Larger oil globules rise more quickly to the surface.
    6. Suspended solids settle and slide down the upper surface of the plates.
    7. The separated oil accumulates at the top and is removed.
    8. Clarified water flows through a final separation chamber and exits the unit.

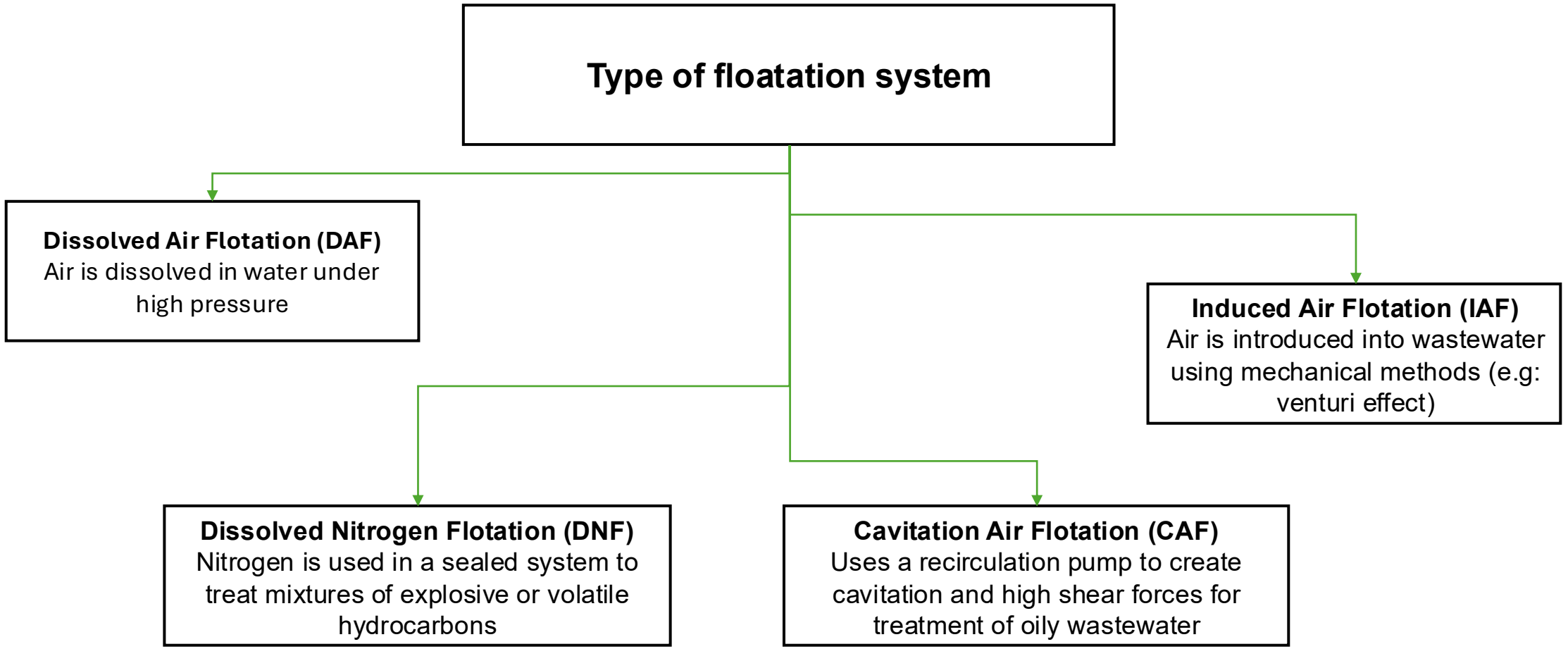
CHEMICAL COAGULATION & FLOCCULATION

- Chemical coagulation and flocculation are used to improve gravity oil separation, especially when wastewater contains:
    1. Emulsified oil
    2. Colloidal particles
    3. Fine suspended solids
    4. Stable oil-water mixtures that cannot be separated by gravity alone
- The process uses chemicals to destabilize emulsions and allow small particles and oil droplets to combine into larger flocs that can be removed by settling or flotation.

CHEMICAL COAGULATION & FLOCCULATION

- Working Principle of Coagulation and Flocculation:
- Coagulation:
    1. Rapid mixing of chemicals (example: alum,ferric chloride,Polyaluminium Chloride (PAC)) into wastewater.
    2. Neutralizes negative charges on colloidal particles.
    3. Destabilizes emulsified oil and fine suspended solids.
    4. Breaks stability of particles so they can start combining.
- Flocculation:
    1. Slow and gentle mixing after coagulation.
    2. Promotes collision between destabilized particles.
    3. Forms larger particles called flocs.
    4. Increases particle size and weight for easier separation.
    5. Prepares flocs for removal by sedimentation or flotation (example: dissolve air floatation (DAF)).

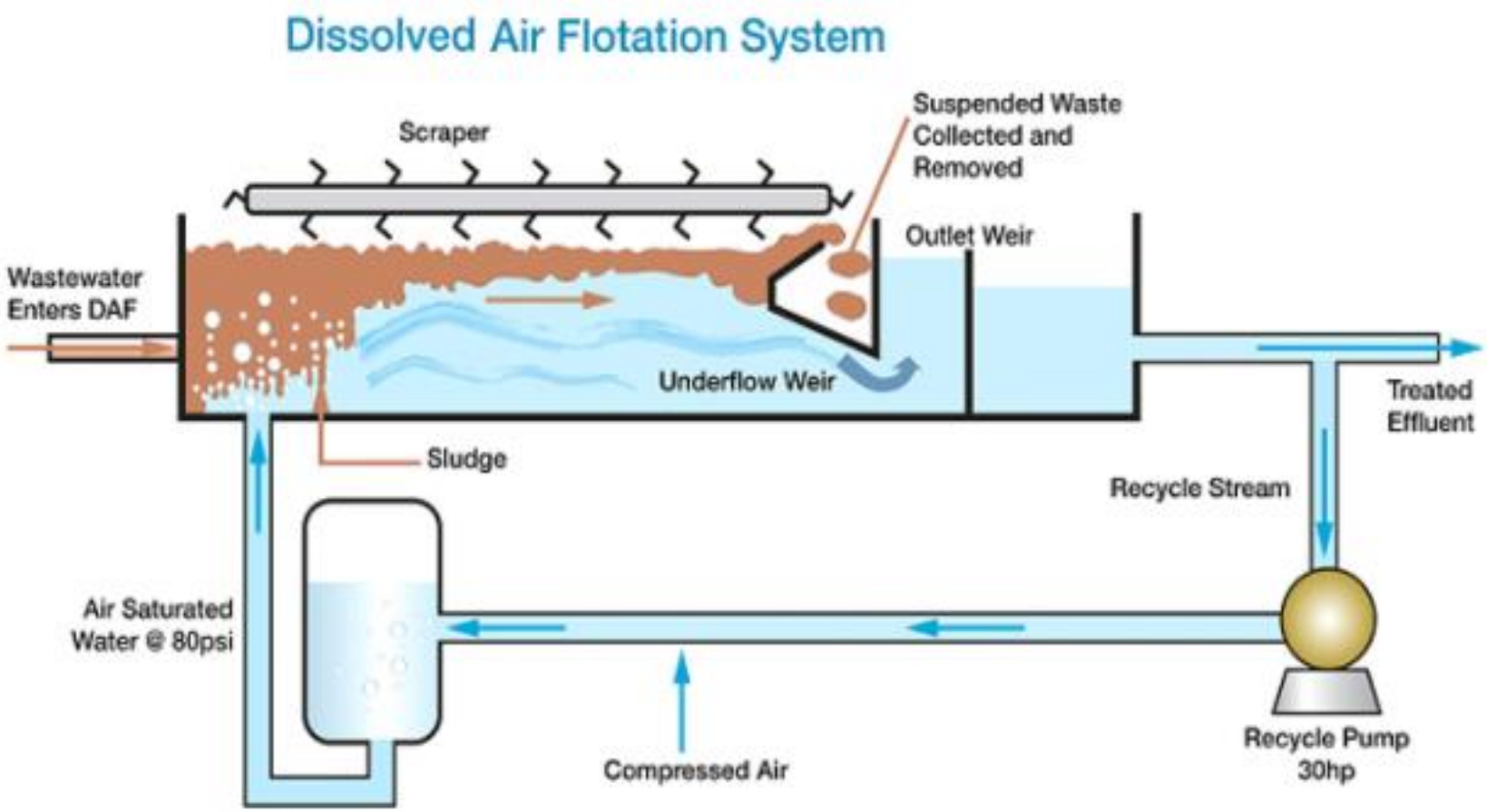
AIR FLOATATION



DISSOLVE AIR FLOATATION

- Dissolved Air Flotation (DAF) is a physical separation process commonly used in wastewater treatment to remove fats, oil, grease (FOG), and fine suspended solids, especially after chemical coagulation.
- Working Principle of Dissolved Air Flotation (DAF):
    1. Wastewater is first pre-treated (often coagulation and flocculation) to form small flocs.
    2. Air is dissolved into water under high pressure.
    3. The pressurized air-water mixture is released into a flotation tank at atmospheric pressure.
    4. Fine microbubbles are formed when pressure is suddenly reduced.
    5. Microbubbles attach to suspended solids, oil, grease, and flocs.
    6. Attached bubbles reduce the density of particles, making them buoyant.
    7. Particles float to the surface and form a sludge (float layer).
    8. Surface sludge is removed using skimmers.
    9. Clarified water is collected from the bottom outlet.

DISSOLVE AIR FLOATATION



EXAMPLE 5

1. Explain 3 types of floatation system.

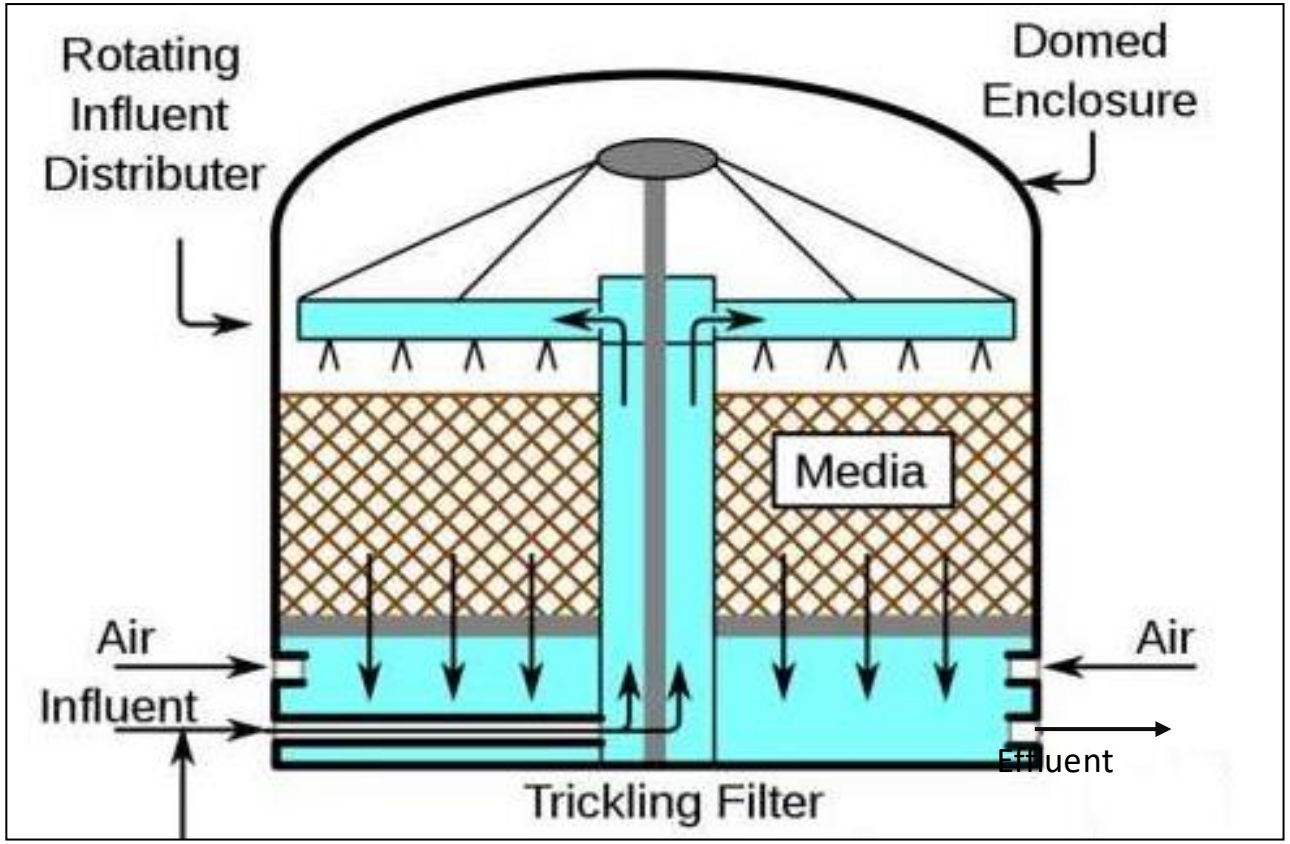
2. Explain the flocculation process.

TRICKLING FILTER (BIOLOGICAL FILTRATION).

- A trickling filter is an attached-growth biological wastewater treatment process where wastewater is distributed over a bed of media, and microorganisms growing on the media surface remove organic pollutants.
- Despite its name, it is not a true filtration process. Instead, treatment occurs through biological activity on a slime layer (biofilm) covering the media.
- Working Principles of trickling filter:

1. Wastewater is sprayed over a bed of packing media using a rotating distribution arm
2. The media is covered with a biological slime (biofilm)
3. As wastewater flows over the biofilm, microorganisms will adsorb organic matter and consume pollutants (mainly BOD)
4. Treated water flows downward and is collected at the bottom
5. The system is supplied with oxygen through air spaces between the media (aerobic conditions maintained)

TRICKLING FILTER (BIOLOGICAL FILTRATION).







NUTRIENT REMOVAL (NITROGEN & PHOSPHORUS)

- Nutrient removal is an advanced wastewater treatment process used to remove excessive nitrogen (N) and phosphorus (P) from wastewater before discharge into rivers, lakes, or coastal waters.
- Excess nutrients can stimulate excessive algal growth, leading to eutrophication, oxygen depletion, fish kills, and deterioration of water quality.
- Removal techniques:

Nutrient	Removal Method	Process
Nitrogen	Biological	• Convert toxic ammonia into nitrate (Nitrification) • Nitrate is converted into nitrogen gas (Denitrification)
Phosphorus	Biological	• Phosphorus Accumulating Organisms (PAOs) absorb phosphorus and stored inside microbial cells • It produce biomass as sludge and phosphorus leaves the system together with the sludge.
Phosphorus	Chemical	• Precipitation using alum or ferric salts

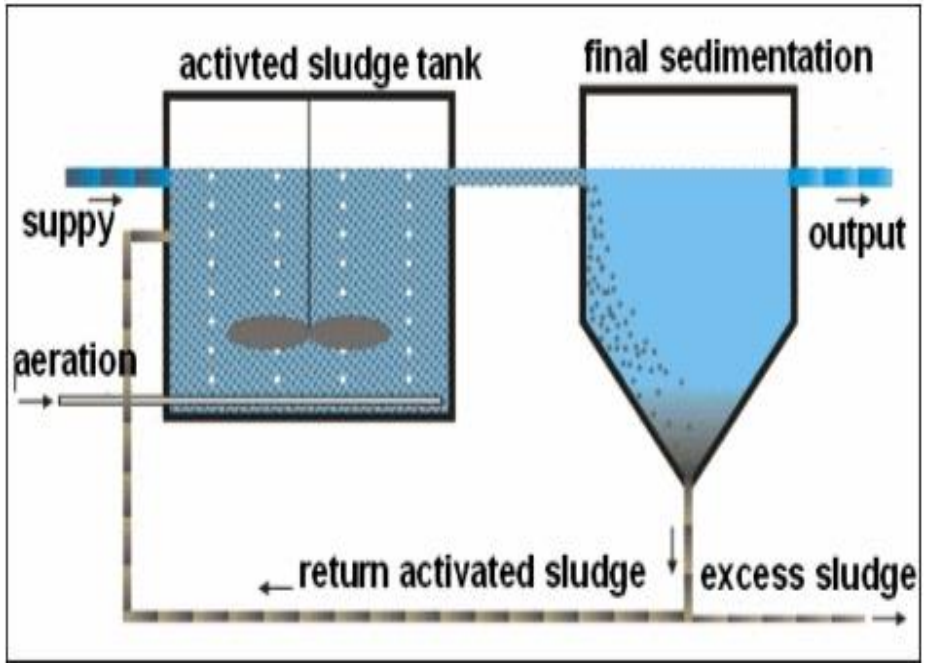
ACTIVATED SLUDGE PROCESS

- The activated sludge process is the most widely used suspended-growth biological wastewater treatment process. In this system, microorganisms remain suspended in wastewater and utilize organic pollutants as food under aerobic conditions.
- Process aims:

1. Remove dissolved and suspended organic matter (BOD)
2. Reduce pollutant concentration in wastewater
3. Produce a high-quality treated effluent
4. Accelerate biological degradation through aerobic microorganisms

ACTIVATED SLUDGE PROCESS

- Working Principle of activated sludge generation process :



1. Wastewater enters the aeration tank
2. Air or oxygen is supplied through diffusers
3. Oxygen maintains aerobic conditions for microbial growth
4. Microorganisms consume organic pollutants as a food source
5. Aeration keeps microorganisms and wastewater thoroughly mixed
6. The mixture of wastewater and microorganisms is called mixed liquor
7. Mixed liquor flows to a secondary clarifier
8. Microbial flocs settle to the bottom
9. A portion of the settled sludge is returned to the aeration tank as Return Activated Sludge (RAS)
10. Excess sludge is removed as Waste Activated Sludge (WAS)
11. Treated effluent leaves the clarifier for further treatment or discharge

OXIDATION POND

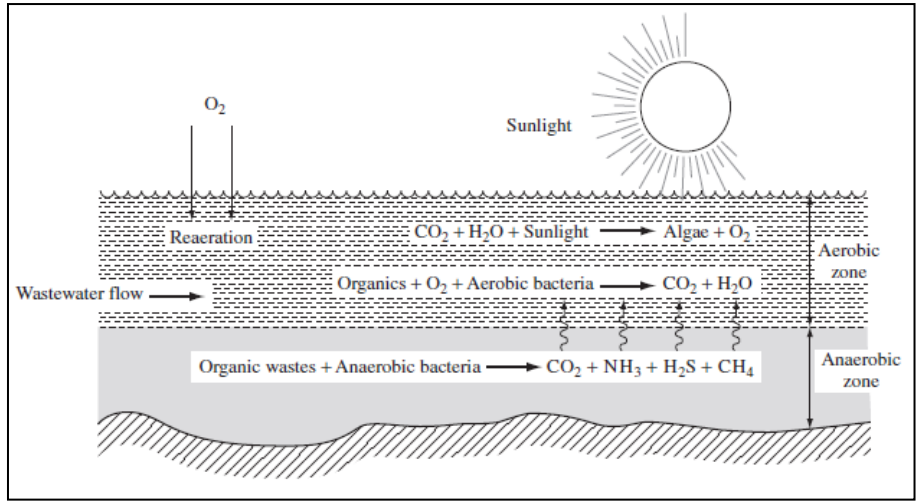
- An oxidation pond is a large, shallow wastewater treatment pond used to treat raw or partially treated sewage through natural biological processes. Microorganisms decompose organic matter in the wastewater, while oxygen is supplied through natural aeration and algal photosynthesis.
- Process aims:

1. Remove organic pollutants (BOD)
2. Stabilize wastewater using natural biological processes
3. Reduce suspended solids and pathogens
4. Provide low-cost wastewater treatment

OXIDATION POND*

Working Principle of oxidation pond:

1. Wastewater enters a shallow pond (typically 1 to 2 m deep)
2. Organic matter is decomposed by microorganisms
3. Oxygen is supplied by:



4. Aerobic bacteria degrade organic matter in oxygen-rich zones
5. Anaerobic bacteria decompose organic matter in oxygen-deficient zones
6. Treated effluent is discharged after sufficient retention time



OXIDATION POND

Type of oxidation pond:

Type	Characteristics
Aerobic Pond	• Oxygen present throughout the pond • Decomposition mainly by aerobic bacteria
Anaerobic Pond	• No dissolved oxygen • Decomposition by anaerobic bacteria
Facultative Pond	• Upper layer undergo aerobic decomposition and algal photosynthesis • Lower layer will undergo sludge digestion and anaerobic decomposition

EXAMPLE 6

1. Explain trickling filter function.

2. State 3 types of oxidation pond.

ATTACH GROWTH TREATMENT VS SUSPENDED GROWTH TREATMENT

- Wastewater biological treatment systems are mainly classified into Attached Growth (Fixed Film) and Suspended Growth processes.
- The difference lies in where microorganisms grow and how they interact with wastewater to remove organic pollutants.

	Attached Growth Treatment (AGT)	Suspended Growth Treatment (SGT)
Definition	Microorganisms grow attached to a surface (media) forming a biofilm	Microorganisms are freely suspended in wastewater
Biomass location	Fixed on packing media	Mixed throughout liquid (mixed liquor)
Examples of process or equipment	1. Trickling filter 2. Rotating Biological Contactor (RBC) 3. Biofilter	1. Activated sludge process 2. Aerated lagoon & oxidation pond
Microbes application	Wastewater flows over biofilm	Continuous mixing of wastewater and microbes
Sludge handling	Biofilm periodically sloughs off	Sludge settles in secondary clarifier and recycled (RAS)
BOD removal	Moderate BOD removal	High BOD removal efficiency

EXAMPLE 7

- Compare both AGT and SGT approaches in terms of the application of microorganisms for further removal of BOD and suspended solids in wastewater.

THANK YOU

PLEASE SUBMIT YOUR MIND MAP